Facial Feature Extraction Using Geometric Feature and Independent Component Analysis

Toan Thanh Do[1] and Thai Hoang Le[2]

[1] Department of Computer Sciences, University of Natural Sciences, HCMC, Vietnam
dttoan@fit.hcmuns.edu.vn
[2] Department of Computer Sciences, University of Natural Sciences, HCMC, Vietnam
lhthai@fit.hcmuns.edu.vn

Abstract. Automatic facial feature extraction is one of the most important and attempted problems in computer vision. It is a necessary step in face recognition, facial image compression. There are many methods have been proposed in the literature for the facial feature extraction task. However, all of them have still disadvantage such as not complete reflection about face structure, face texture. Therefore, a combination of different feature extraction methods which can integrate the complementary information should lead to improve the efficiency of feature extraction stage. In this paper we describe a methodology for improving the efficiency of feature extraction stage based on the association of two methods: geometric feature based method and Independent Component Analysis (ICA) method. Comparison of two methods of facial feature extraction: geometric feature based method combined with PCA method (called GPCA) versus geometric feature based method combined with ICA method (called GICA) on CalTech dataset has demonstrated the efficiency of GICA method. Our results show that GICA achieved good performance 96.57% compared to 94.70% of GPCA method. Furthermore, we compare two methods mentioned above on our dataset, with performance of GICA being 98.94% better 96.78% of GPCA method. The experiment results have confirmed the benefits of the association geometric feature based method and ICA method in facial feature extraction.

Keywords: Face recognition; independent component analysis (ICA); principal component analysis (PCA); geometric features.

1 Introduction

Face recognition has a variety of potential applications in public security, law enforcement and commerce such as identity authentication for credit card or driver license, access control, information security, and video surveillance, etc.. In addition, there are many emerging fields that can benefit from face recognition such as human-computer interfaces and e-services, including e-home, tele-shopping and tele-banking.

However, one of the most difficult from face recognition problem is the facial feature extraction step. A good feature extraction will increase performance of face recognition system. Various techniques have been proposed in the literature for this purpose, and are mainly classified in four groups.

D. Richards and B.-H. Kang (Eds.): PKAW 2008, LNAI 5465, pp. 231–241, 2009.
© Springer-Verlag Berlin Heidelberg 2009

1.1 Geometric Feature Based Method Group

The features are extracted by using relative positions and sizes of the important components face such as eyes, nose, mouth and other important component face. Almost this group concentrates in two directions.

Firstly, detecting edges, directions of important components or region images contain important components, then building feature vectors from these edges and directions. Using filters such as Canny filter to detect eyes or/and mouth image region, using transforms method such as Hough transform to detect eyes [1] and the gradient analysis is a method which is usually applied in this direction.

Secondly, methods are based on the grayscales difference of important components and unimportant components, by using feature blocks, set of Haar-like feature block [2] in Adaboost method to change the grayscales distribution into the feature vector. One of the proposed methods is LBP method [4]; it divides up the face image to regions (blocks); each region corresponds with each central pixel; we examine its pixel neighbors, based on the grayscales value of central pixel to change its neighbor to 0 or 1. Therefore, every pixel will be represented in a binary string. Since then, we build histograms for every region. Then these histograms are combined to a feature vector for the face image. The two methods, Cascaded AdaBoost [3] and Chain AdaBoost, use a set of Haar-like features to detect important components of face or/and face. Selecting features is based on different grayscales or "integral image" in every Haar-like feature block. In Gabor Wavelets method [5], Gabor filter can capture salient visual properties such as spatial localization, orientation selectivity, and spatial frequency characteristics. The advantages of these methods are concentration on important components of face such as eyes, nose, mouth, etc. but the disadvantage is not to remain face global structure and face texture.

1.2 Template Based Method Group

Based on a template function and appropriate energy function, this method group will extract feature of important component of face such as eyes, mouth, etc. or face shape.

An image region is the best appropriateness with template (eyes, mouth, etc.) which will make minimize the energy. The methods have been proposed such as deformable template and genetic algorithms. In the deformable template method [6] [7] [8], the feature of interest, an eye, for example, is described by a parameterized template. An energy function is defined to links edges, peaks, and valleys in the image intensity with corresponding properties of the template. Then the template interacts dynamically with the image, by altering its parameter values to minimize the energy function, thereby deforming itself to find the best fit. The final parameter values can be used as descriptors for the features. Advantages of this group method are using template and determining parameter for important components of face, but disadvantage is not to reflect face global structure.

1.3 Color Segmentation Based Method Group

This group method based on skin's color to isolate the face .Any non-skin color region within the face image region is viewed as a candidate for "eyes" and/or "mouth" [9] [10].